Flight Simulator

October University for Modern Science and Arts
(MSA University)

Faculty of Engineering
Department of Mechatronics

By: Ahmad Essameldin

Flight Simulator

A graduation project 2023

By:

Ahmad Essameldin Ahmad

Omar Ashraf Abuzaid

Under the supervision:

Prof. Dr. Ahmad Badawy

Assoc. Prof. Dr. Amgad Bayoumy

Problem Statement

- Commercial flight simulators have mostly been tailored for professional, commercial airline certified users.

THE PROBLEM

High Cost of Commercially Available Flight Simulators/ Motion Platforms.

Flight Simulators On The Market Also Suffer From Being Too Proprietary/ Not Universal In Nature

Problem Statement

THE PROBLEM

High Cost of Commercially Available Flight Simulators/ Motion Platforms.

	Fidelity Level	DOF	FOV	Price
The Boeing 737 NG	High	Six	180-degree	$10-25 (million)
Airbus A320	High	Six	200-degree	$10-25 (million)
CAE 7000XR Series	High	Six	220-degree	$10-25 (million)
The Redbird	Low	Three	210-degree	$150-250 (thousands)

PROJECT Importance

Project Aim

To undertake the design and hardware implementation of a human-in-the-loop flight simulator complete with peripherals and a motion platform

Circumvent the excessive costs associated with purchasing a platform with similar features.

THE PROBLEM

PROJECT Prototype

Project Results

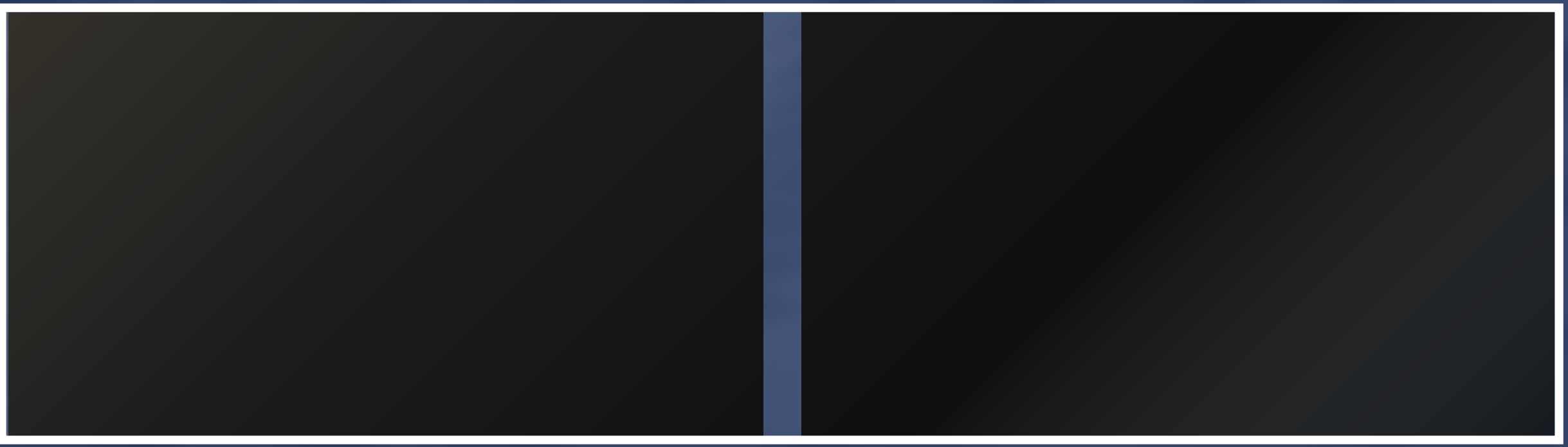
Project Aim

Circumvent the excessive costs associated with purchasing a platform with similar features.

01/04/2023		Laser - Sheet metal 2.5mm - Electrostatic Painting)	Factory	2200
02/04/2023		10 mm drill bit - meter - spray -	Aly-eldeen	275
26/04/2023		Fan - T nuts - Wire - desoldering Pumpm	HD - Electronics	195
30/04/2023		Endcaps - 2wa3ed - heatshrink - terminals - laser cutting - Cables	Aly-eldeen	480
16/05/2023		Motor driver and endcaps	Free Electronics	740
05/04/2023		Rotary encoders+Precision potentiometers	Free Electronics	220
08/05/2023		USB Extension+Power Cable+jumpers+DC convert[illegible] wire	Circuits	300
07/05/2023		10A dual chan[illegible] mot[illegible]	[illegible]	5[illegible]
22/05/2023		Pliers + Circ[illegible] bre[illegible]er	[illegible]	1[illegible]
22/05/2023		Springs - Dri[illegible]its - Sp[illegible]	[illegible]	2[illegible]
05/06/2023		Springs	Sabteya	140
			TOTAL Cost	17039

Total Cost: 22000EGP

PROJECT Prototype



Fixed Platform

2 DOF Motion Platform

Production & offered features

BASIC

EGP **14999** 99

Fixed Platform

- ✓ Pilot Ergonomics
- ✓ Monitors, Speakers
- ✓ Yoke, Pedals, Switch Board
- ✓ Cockpit Customizations
- ✓ Virtual Reality
- ✗ Haptic Feedback
- ✗ Motion Feedback

Standard

EGP **44,999** 99

2 DOF Motion Platform

- ✓ Pilot Ergonomics
- ✓ Monitors, Speakers
- ✓ Yoke, Pedals, Switch Board
- ✗ Cockpit Customizations
- ✓ Virtual Reality
- ✓ Haptic Feedback
- ✓ Motion Feedback

Premium

EGP **89,999** 99

6 DOF Motion Platform

- ✓ Pilot Ergonomics
- ✓ Monitors, Speakers
- ✓ Yoke, Pedals, Switch Board
- ✓ Cockpit Customizations
- ✓ Virtual Reality
- ✓ Haptic Feedback
- ✓ Motion Feedback

This is a preliminary estimation for the cost and doesn't include the already purchased components: Like: PC, Monitors, ..

Production & offered features

Target Customers

Military Pilot Trainings

Aerospace Academies and Universities

Mission planning and Rehearsal

Flight Schools and Training Centers

Research Centers

Entertainment for Aviation Enthusiasts

Thank You